

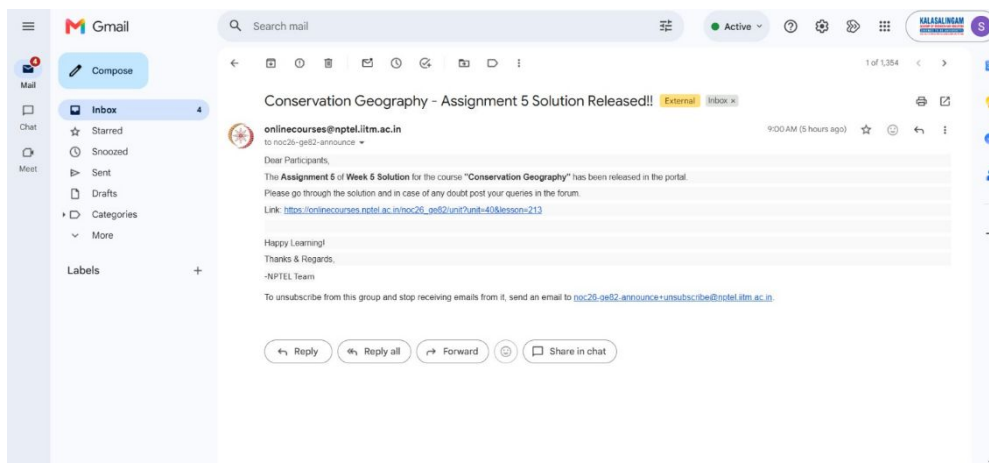
Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

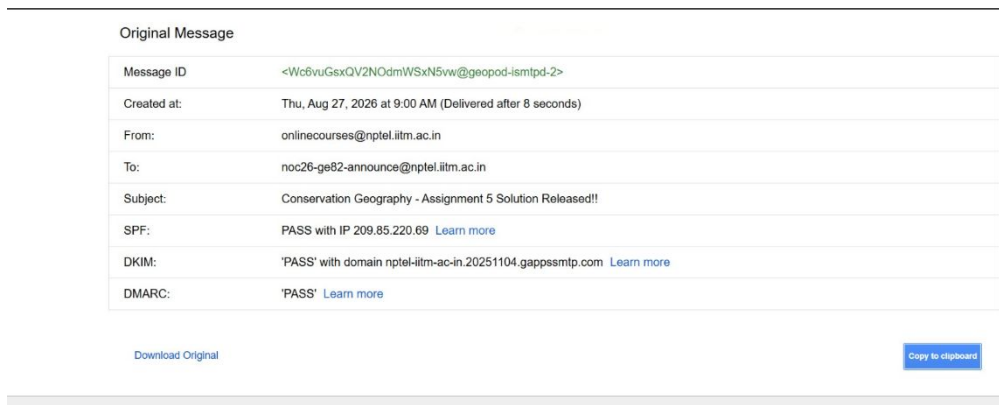
1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."



2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.



3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - The sending server's hostname/IP address.
 - The receiving server's hostname/IP address.
 - The date and time the email passed through.

The screenshot shows the MXToolbox 'Analyze Headers' tool interface. At the top, there's a navigation bar with links like Pricing, Tools, Delivery Center, Monitoring, Products, Resources, and Login. Below this is a dark navigation bar with various tool categories, and 'Analyze Headers' is selected. The main content area shows 'Header Analyzed' for the email subject 'Conservation Geography - Assignment 5 Solution Released!!'. A yellow warning box states: 'Copy/Paste Warning: Copy/Pasting a header works for most people, but sometimes it can cause problems with things like DKIM Validation. For the best results, use our Email Deliverability tool'. Under 'Delivery Information', there's a list of checks: DMARC Compliant (green), SPF Alignment (green), SPF Authenticated (green), DKIM Alignment (red), and DKIM Authenticated (green). Under 'Relay Information', a table shows 'Received Delay:' as '7 seconds'.

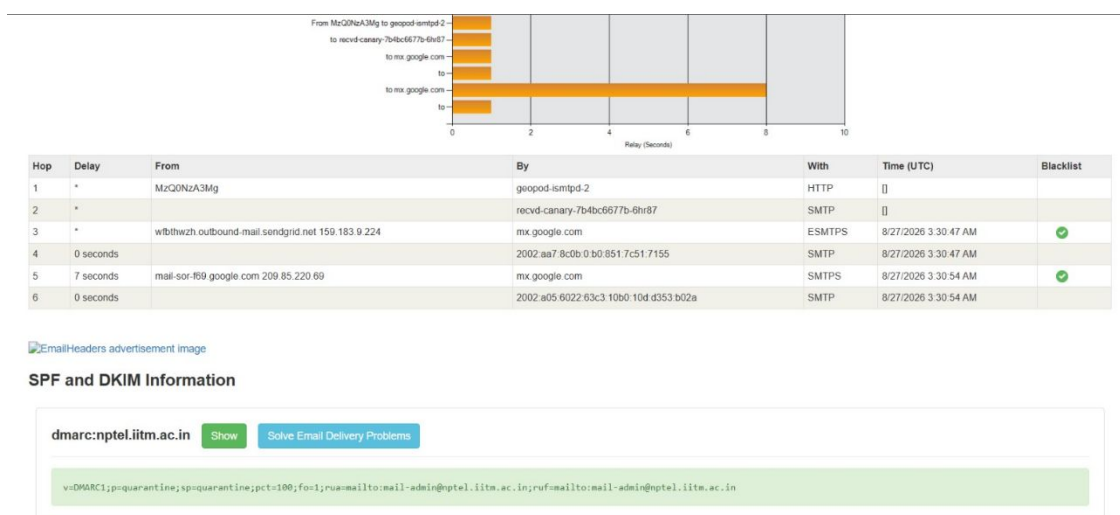
5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.

- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - o **Pass:** Indicates the IP is authorized.
 - o **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - o **Pass:** The email has not been tampered with.
 - o **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - o **Pass:** Indicates alignment with SPF/DKIM policies.
 - o **Fail:** Indicates possible spoofing or tampering.



7. Analyze the 'Message-ID'

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in ReturnPath, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.

Headers Found

Header Name	Header Value
Delivered-To	9924004122@klu.ac.in
X-Forwarded-Encrypted	i=4; AHgh+RoB2j+VqQ21NPTveqrYmVXoHvD1yIqggh3qfP07B0ARKkzJSXcEON8mZn7fZURFTAd11UhaMTAniQw==@klu.ac.in
X-Received	by 2002:a05:6a00:8d86:b0:848:2eac:bfb2 with SMTP id d2e1a72fcca58-853748a0551m/2397447063a.13.1787801454456, Wed, 26 Aug 2023 20:30:54 -0700 (PDT)
ARC-Seal	i=3; a=rsa-sha256, t=1787801454, cv=pass; d=google.com; s=arc-20260327; b=FxHvdukeN1xNpq4NH8LmIT3VZNeK11NVL4i2Mws0NoOHLfrXJRk0ba50aE5G 1MghHnBRLTW8Vzqt5iObb8aPah6DSQHngGSsi7zyLS NFjHlALegV32U2jJOJSMt9 JT1De1nkYgdmMyODIRI77ZxrtYMuCgmMhFhDAV++8uNvryq7T4asPetrG0d9fCEq8 30uVtaxbhwZCR1p0R2G29LGAZOpWVGPhA0bOcdogmu8AcR4a04gVofCw/aSove rvz9W9MsZ9/Tnr Y8GzCvndqxyKQk+7W/CqCcbobXh+XK0sCDJ8xtY0tYgEvk6lZc0xv W7Rg==
ARC-Message-Signature	i=3; a=rsa-sha256, c=relaxed/relaxed, d=google.com, s=arc-20260327, h=lists-unsubscribe-list-archive-list-help-list-post-list-id-mailing-list-precedence-to-subject-message-id-mime-version-from-date-content-transfer-encoding-dkim-signature, bh=2FFQgMbczd43JUTVeD0TdfWAeNUi4dwqHRQlyz4ns=, fh=PRJJB71obMdaDlIf4O6dnj8F9dtSNwHlab5CuCW8=, b=Rxbdu8+khqihY0Jbc3ql776eX7rm3TswukTJlA/Z+rwUjYm8X8dLZphvZulvO 5Fo240Ghwcpu5QqrUxxNvSu9mD+XIQOpk4BPY/vQ1jBSq6lJUSEKDbKNtlyqcsWsq OvOvkS31TrvcexT53xr7otBaNE8DrGpoohtuadYGgp0BDcuEo49eA4ELz06NjU3q uH9s9Pw9fk08j0avcx0thY7cRyAdiqdV0gXdCZ qNemwE0t/sxnUqSBu05Ei0vXsd yfJyWhzHEomWAZHf6KsknC1U6UINxsVpqn9fhdByWpdDoGhyGL15BFvCsP5uYj1L OK+A==; data=google.com
ARC-Authentication-Results	i=3; mx.google.com; dkim=pass header=i=@ntel-itm.ac.in 20251104 gappssmtp.com header=s=20251104 header=b=T9MQS3h; arc=pass (i=2 spf=pass spfdomain=mail.ntel.itm.ac.in dkim=pass dkdomain=ntel.itm.ac.in dmarc=pass fromdomain=ntel.itm.ac.in); spf=pass (google.com; domain=ntel.itm.ac.in); dmarc=pass (p=QUARANTINE sp=QUARANTINE ds=NONE) header=from=ntel.itm.ac.in; dara=neutral header=i=@klu.ac.in
Return-Path	<ntel-itm.ac.in>
Received-SPF	pass (google.com; domain=ntel.itm.ac.in); ip=209.85.220.69 as permitted sender) client-ip=209.85.220.69;
Authentication-Results	mx.google.com; dkim=pass header=i=@ntel-itm.ac.in 20251104 gappssmtp.com header=s=20251104 header=b=T9MQS3h; arc=pass (i=2 spf=pass spfdomain=mail.ntel.itm.ac.in dkim=pass dkdomain=ntel.itm.ac.in dmarc=pass fromdomain=ntel.itm.ac.in); spf=pass (google.com; domain=ntel.itm.ac.in); dmarc=pass (p=QUARANTINE sp=QUARANTINE ds=NONE) header=from=ntel.itm.ac.in; dara=neutral header=i=@klu.ac.in
DKIM-Signature	v=1; a=rsa-sha256, c=relaxed/relaxed, d=ntel-itm.ac.in 20251104 gappssmtp.com; s=20251104; t=1787801454; x=1788406254; dm=klu.ac.in; h=lists-unsubscribe-list-archive-list-help-list-post-list-id-mailing-list-precedence-to-subject-message-id-mime-version-from-date-content-transfer-encoding; from=ntel-itm.ac.in; to=cc:subject:date:message-id:reply-to:content-type; bh=2FFQgMbczd43JUTVeD0TdfWAeNUi4dwqHRQlyz4ns=; b=T9MQS3hPIUOYFyLYnzK5oglt+WGxLLgBV3iWDGFXXq+OS0AJNaMqAERTEdi CvFwlsQyTn8Fk6Gdzunx08k3j7RyTL6wk7mY1wUjacrQ4UWZ/CTXKikR0YjpeEthj Q0H3ubcuP4 FNMerrtGxL/RkH1PH6043is2IG80zqTk46ASQLcu08KKCOewYZI gW4UBFaBIO/WetxRBMORZif6jv9+O+Ju6nqSnrRdV28Df8j8m504F1jEBvSrdG+b JbxtVfXAZx9xowmREkvy/8K0dkzQ9/GJ/Snp+v9e03yudYx1h PTISHICl7ILB2NK7z h9fg==
X-Google-DKIM-Signature	v=1; a=rsa-sha256, c=relaxed/relaxed, d=1e100.net, s=20251104, t=1787801454, x=1788406254, h=lists-unsubscribe-list-archive-list-help-list-post-x-spam-checked-in-group-list-id-mailing-list-precedence-x-original-authentication-results-x-original-sender-to-subject-message-id-mime-version-from-date-content-type-content-transfer-encoding-x-beenthere-x-gm-message-state-from-to-cc:subject:date:message-id:reply-to:content-type, bh=2FFQgMbczd43JUTVeD0TdfWAeNUi4dwqHRQlyz4ns=; b=T9MQS3hPIUOYFyLYnzK5oglt+WGxLLgBV3iWDGFXXq+OS0AJNaMqAERTEdi CvFwlsQyTn8Fk6Gdzunx08k3j7RyTL6wk7mY1wUjacrQ4UWZ/CTXKikR0YjpeEthj Q0H3ubcuP4 FNMerrtGxL/RkH1PH6043is2IG80zqTk46ASQLcu08KKCOewYZI gW4UBFaBIO/WetxRBMORZif6jv9+O+Ju6nqSnrRdV28Df8j8m504F1jEBvSrdG+b JbxtVfXAZx9xowmREkvy/8K0dkzQ9/GJ/Snp+v9e03yudYx1h PTISHICl7ILB2NK7z h9fg==

Example of a Header Analysis

Here's a quick example with a sample header:

vbnet

Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1]) by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

Received: by mail.example.com with SMTP id a1mr1243772ywh.51;

Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.

